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Development Project

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PROJECT PROPOSAL

Aqua Mind

AI-Powered Smart Aquarium Management System

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AQUA MIND

AI-POWERED SMART AQUARIUM MANAGEMENT SYSTEM

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Acknowledgment

It is with utmost sincerity that I like to extend my deepest appreciation to all the people who have contributed their efforts and supported me in the process of developing this AquaMind – AI-Powered Smart Aquarium Management System project proposal.

First, I would like to acknowledge the efforts and guidance provided by my lecturers and academic advisors in helping me conduct this project and giving me constructive criticism and ideas for this research. In addition to the above, I would like to thank all my friends and classmates for their efforts and motivations during the process of preparing for this project. Lastly, I would like to extend my deepest appreciation to my family for their encouragement and motivation in making this project successful.

Executive Summary

AquaMind is a smart aquarium management system that utilizes artificial intelligence capabilities to enhance the process of managing an aquarium by monitoring, maintaining, and supporting decision making in a smart way. AquaMind is implemented as a website where users can effectively manage multiple aquariums by tracking the status of their water, types of fish and plants living there, and organizing various maintenance operations.

This project implements advanced functionality using Artificial Intelligence (AI) and Machine Learning (ML) including recommendation systems for fish and plants, predicting water quality, providing smart notifications, as well as presenting interactive data on dashboard. AquaMind will solve such common problems for aquarium owners as improper water quality, incompatible combination of fish and plants, and missing maintenance procedures.

React, Node.js with Express.js, Python, and Microsoft SQL Server are selected as the key technologies to be used to implement this application. Feasibility study indicates that the project is feasible both from technical, economic, operational, and legal perspectives. The main objective of AquaMind is to offer efficient, convenient, and intelligent solutions for aquarium health management.

Title

Aqua Mind – AI-Powered Smart Aquarium Management System

Introduction

Maintaining an aquarium has become one of the most fascinating and exciting hobbies practiced worldwide by millions of people. Yet, a lot of work must be done to maintain the appropriate aquarium environment since the process entails constant monitoring of water parameters, correct fish and plant selection, regular aquarium maintenance, and timely decision making. Often, it is hard for amateur aquarists to deal with the abovementioned tasks in a proper way, resulting in water contamination, unhealthy living conditions for their aquatic pets, and elevated levels of fish mortality. Furthermore, existing approaches for aquariums' monitoring and management are inefficient, time-consuming, and not as accurate as one would want them to be.

We propose to implement Aqua Mind – AI-Powered Smart Aquarium Management System as a web-based platform to facilitate and optimize the process of aquarium maintenance. Using state-of-the-art web development solutions in combination with artificial intelligence (AI) and machine learning (ML), Aqua Mind will help users in managing their aquaria better and more efficiently.

Our product will give users opportunities for maintaining multiple aquaria, monitoring the water parameters, keeping record of all relevant information regarding fish and plants, scheduling aquarium maintenance, and analyzing the performance of the aquariums using the interactive dashboard. Moreover, with the help of our recommendation engine, users will receive recommendations for selecting fish and plants depending on conditions of a given aquarium. Finally, our prediction module will make it possible to predict the future state of the water in the tank.

Current Situation Leading to Problem Identification

People who have aquariums in Sri Lanka use methods to take care of their aquariums. They use methods, simple mobile applications, and international community forums. This way of doing things is not particularly good because it has gaps that affect the health of the aquarium and the experience of the aquarium hobbyists. Aquarium hobbyists must deal with these problems every day. Aquarium hobbyists need a way to manage their aquariums.

1. **Manual Tracking of Aquarium Data**

People usually write down water details like pH and temperature by hand in notebooks or spreadsheets. This takes a lot of time and is hard to keep up with, and mistakes can happen easily. If you do not have an effective way to keep track of what happened in the past and can't see the information in an effortless way, it is tough to spot patterns or predict when water quality might become a problem.

2. **Lack of Intelligent Aquarium Management**

Existing applications focus on simple logging and reminders. They do not provide AI-based recommendations for fish compatibility, plant selection, or preventive maintenance. Users must rely on personal knowledge or internet forums for decision-making.

3. **No Predictive Water Quality Monitoring**

Most water systems today are not equipped to predict problems before they happen. This means that people usually only act after something has gone wrong, instead of stopping it from happening in the first place. When the water quality is bad, it can cause a lot of stress for the fish, make them sick, and even kill them.

4. **Fragmented Multi-Tank Management**

For people who have lots of aquariums, keeping track of everything can be a challenge. They usually must use different tools and apps to monitor each tank, which can be frustrating. What is really needed is a single platform where they can see everything that is going on with all their tanks at the same time.

5. **Poor Maintenance Scheduling**

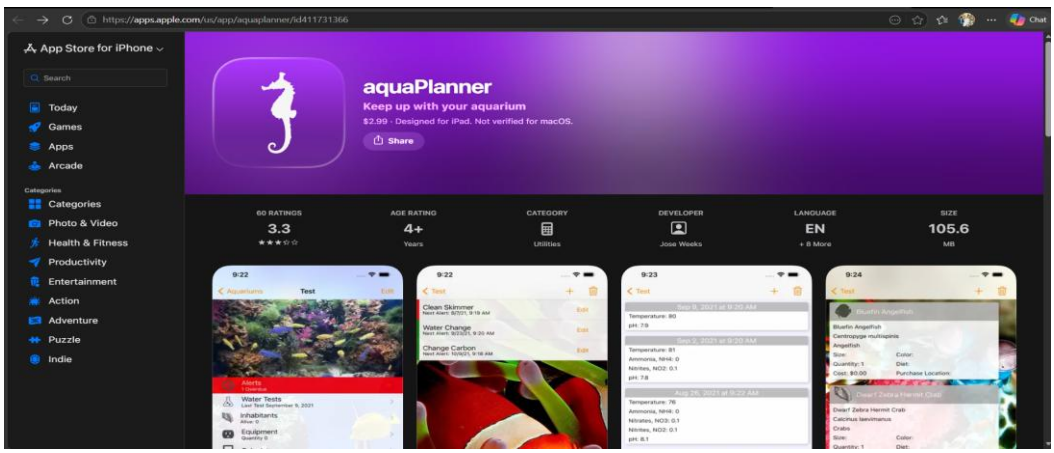
Important maintenance tasks such as water changes, filter cleaning, and feeding schedules are frequently forgotten due to lack of automated reminders and intelligent scheduling systems.

Existing System Comparison

1. **Aquaplanner**

   Aquaplanner is a mobile-based aquarium management application mainly designed for iOS users. It helps aquarium hobbyists organize and manage their tanks through reminders and logging features. **(App Store, 2017)**

   - aquaPlanner App - App Store



Key Features

- Water parameter logging
- Maintenance reminders
- Multi-tank management
- Equipment tracking
- Data export functionality

Strengths

The application is simple and easy to use for people who are just starting out. The application lets users take care of aquariums and it reminds us of regular maintenance tasks such as water changes and feeding schedules.

Limitations

Even though AquaPlanner has some features that help users manage their aquariums it does not have any special features like Artificial Intelligence that could help users make better decisions. The system cannot look at the information that users put in and suggest what kind of fish would be good for their aquarium or warn them about problems with the water quality in the future.

2. Aquadayum

Aquadayum is a software application that uses artificial intelligence technology for managing aquarists' tanks. The software enables aquarium owners to monitor their tanks, identify tank inhabitants, keep track of maintenance tasks, and get advice from the built-in artificial intelligence technology. **(Aquadayum, n.d.)**

https://www.aquadayum.com/



Key Features

- AI-powered species identification
- Aquarium management and monitoring
- Water parameter tracking
- Maintenance scheduling and reminders
- Fish compatibility checking.

Strengths

Aquadayum enables users to have the latest technology with its ability to manage aquariums with advanced AI tools. The platform has intelligent fish recognition, compatibility tools, and aquarium monitoring facilities.

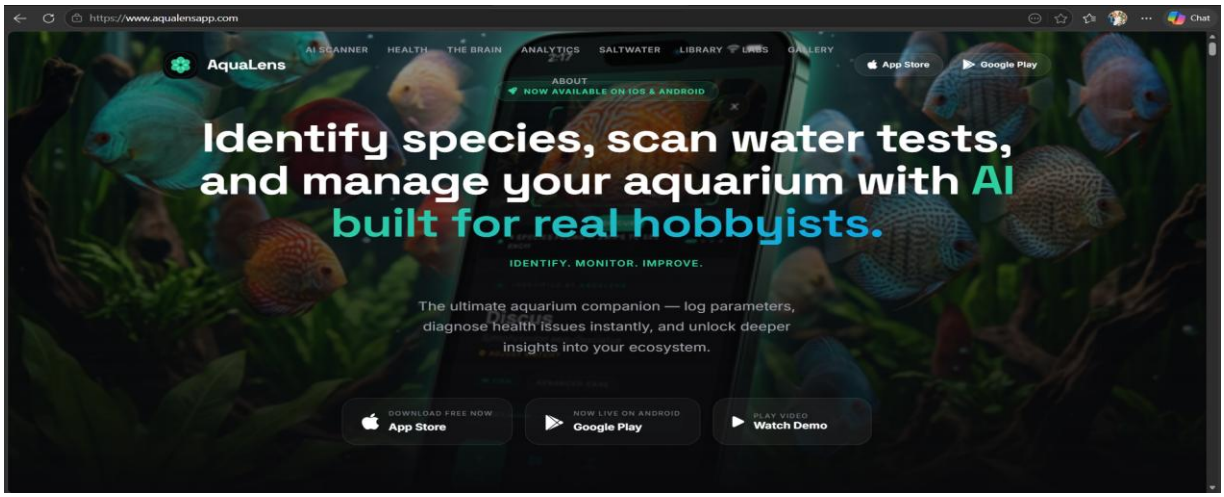
Weaknesses

While Aquadayum offers some AI tools, certain functionalities in the system require internet connectivity and regular updates. Furthermore, while it is designed primarily for aquarium support and monitoring, the system lacks localized predictions for the needs of aquarium owners in Sri Lanka.

3. AquaLens

AquaLens is an AI-powered aquarium companion application designed to help users identify fish species, monitor aquarium conditions, and manage aquarium health using intelligent analysis tools. **(AquaLens, n.d.)**

AquaLens — AI Fish Identifier & Aquarium Management App



Key Features

- AI fish and plant identification
- Water quality monitoring
- Health diagnostics for fish diseases
- Predictive alerts and analytics
- Maintenance tracking
- Species compatibility checking

Strengths

AquaLens has state-of-the-art AI features that enable it to monitor and manage aquarium tanks efficiently. The application comes with advanced analytical capabilities, with support for both freshwater and saltwater aquarium tanks.

Weaknesses

The app heavily relies on AI and connectivity. It also has some features that might be difficult for a beginner user to understand. The system concentrates more on AI analysis and monitoring of the aquarium tanks and not all-around management of multiple tanks.

Overall Analysis Conclusion

These limitations make it evident that there is a necessity for an automated aquarium management system utilizing artificial intelligence (AI). Through integrating existing aquarium management functions and technologies such as artificial intelligence and predictive analysis along with dashboards, the proposed management system can be a smart alternative for aquarium enthusiasts in Sri Lanka.

Proposed Technique to Solve the Current Problem

The AquaMind system suggests a revolutionary web-based system that is infused with intelligent AI capabilities and assists in optimizing the functioning of aquariums using automated procedures and smart decisions. It enables users to easily manage multiple aquariums, measure water quality parameters, control the behavior of fish and plants within the aquariums, and perform various maintenance tasks effectively. Moreover, the system employs AI/ML algorithms to provide smart insights and predictions regarding water quality and alerts users when conditions become dangerous for the health of the aquatic ecosystem.

1. Centralized Multi-Tank Management

The system is designed in such a way that users will be able to control different aquarium tanks using just a single application. It is possible to create different profiles for each aquarium tank, with information on each tank being included in such a profile.

- Tank name and type
- Tank size and dimensions
- Fish and plant species.
- Detail equipment such as filters, heaters, and lighting systems.
- Maintenance history
- Water parameter records

This makes things easy for aquarium enthusiasts by letting them control all their tanks from a single platform rather than using many different applications and manual processes.

2. Water Parameter Logging and Trend Analysis

Water quality is especially important to the survival of fish and aquatic plants in the aquarium. The system gives an option to create a water parameters logger where the user enters critical data on the water quality.

The parameters include:

- pH levels
- Temperature
- Ammonia levels
- Nitrite levels
- Nitrate levels

The system stored there in the database and graphical representation is created based on previous records. This feature helps users make informed decisions and maintain stable aquarium conditions.

3. AI-Based Species Recommendation System

One of the hardest things that aquarists must do is choose fish and plants that complement each other. Incompatibility between the two could result in aggression, stress, poor growth, or even death among the aquatic organisms. The following system will include a recommendation engine using artificial intelligence.

The AI system analyzes several factors such as:

- Tank size
- Water parameters
- Existing fish species
- Tank type (monster tank or planted tank)

Based on these inputs, the AI module suggests:

- Compatible fish species.
- Suitable aquatic plants.
- Recommended stocking levels.

The recommendation engine will enable the user to establish healthier aquarium environments.

4. Machine Learning Water Quality Prediction

The prediction feature transforms aquarium management from a reactive process into a proactive system, improving fish safety and reducing maintenance risks. The prediction model uses historical data about the parameters of water from the aquarium. Through analyzing past trends and patterns, the machine learning model can make predictions regarding:

- Rising ammonia levels
- Decreasing pH levels
- Temperature instability

The prediction capability turns fish keeping into a proactive operation rather than a reactive one.

5. Intelligent Maintenance Scheduling

Maintenance is vital in keeping the aquatic environment healthy and safe. But most aquarium enthusiasts tend to overlook maintenance work because of their hectic schedules. This new system design comprises an intelligent maintenance schedule creator that creates a plan for maintenance activities. The scheduler manages tasks such as:

- Water changes
- Feeding schedules
- Filter cleaning
- Equipment maintenance
- Lighting schedules

The user will be notified about maintenance through the system dashboard. The maintenance activity logs will be saved in the database. The scheduling system helps users:

- Reduce maintenance errors.
- Improve aquarium health.
- Maintain consistent care routines.
- Avoid missed tasks.

It helps the user keep a track record of all maintenance activities for each aquarium.

Feasibility Study

This feasibility study aims to examine the viability of implementing the system based on the resources and technologies available as well as the budgetary considerations involved. In the AquaMind case study, the feasibility study encompasses both technical, financial, operational, and legal considerations for implementation of the proposed solution**. (GeeksforGeeks, 2025)**

Technical Feasibility

Technical feasibility refers to the availability of technology, equipment, and technical skills necessary to implement the system successfully. The AquaMind system is technically feasible since it utilizes advanced and commonly used technologies for website creation, databases, and Artificial Intelligence incorporation.

1. **Frontend :-** React technology was selected for frontend development due to its ability to offer an up-to-date and responsive UI. The framework offers developers the ability to create reusable components for building complex interfaces with ease. It can be used to develop several types of web pages like dashboards, charts, notifications, aquarium monitoring panels, among others.

2. **Backend :-** Node.js with Express.js were used for the backend development due to their ability to support rapid and scalable applications. Node.js is known for its efficiency in processing requests from the browser, communicating with databases, and offering an effortless way of connecting the frontend with the backend in real-time. It is also possible to develop RESTful APIs using express.js.

3. **AI Engine :-** Python was selected as the preferred language for our AI engine development due to its widespread use in developing Artificial Intelligence and Machine Learning applications. It has several powerful libraries that will come in handy when implementing predictive features of water quality and intelligent recommendations, including scikit-learn, pandas, and NumPy.

4. **Database :-** Microsoft SQL Server database management system was chosen for storing the structured data in the database. The database management system will manage storing data on user accounts, aquariums, water parameters history, maintenance schedules, and fish types, among others. It is known for its reliability, efficiency, and security in storing and manipulating structured data.

5. **Communication :-** REST APIs were selected for communication between the components of the software, including the frontend, backend, AI engines, and database. It provides a standardized way of transferring data among various parts of a distributed system. Using REST architecture promotes modularity, scalability, and easy integration of different frameworks and technologies such as React, Node.js, and Python.

Economic Feasibility

Economic viability determines whether the project will be able to develop within an affordable budget and if the project offers enough value for the investment made. This project is economically viable since it uses affordable and free software technology. Most of the development tools that the project will use are freely available without buying costly licenses. The project minimizes cost by using:

- Open-source frameworks and libraries
- Standard personal computer hardware
- Free development environments and tools

This project utilizes affordable technologies including React, Node.js, and Python to lower software development costs. It also will not need costly hardware or Internet of Things sensors during the early stage. Moreover, the system can assist users to save on their overall aquarium maintenance expenses.

Operational Feasibility

The concept of operational feasibility refers to how well the proposed solution can perform its functions in a desired user environment and ease the adoption by users of the proposed solution. In respect to this, the AquaMind solution has been developed to be user-friendly and usable by both beginners and advanced aquarium enthusiasts. The interface of the solution will contain easy navigation, dashboarding, charting, and alerts to improve usability.

The centralized dashboard allows users to:

- Manage multiple aquariums.
- Monitor water conditions .
- Track maintenance schedules.
- View recommendations and analytics.

The system also offers intelligent notification and reminder features, which will enable users to take care of their aquariums correctly without the need for manual monitoring. As the system is online-based, users can gain access to the system through various devices like desktop computers, tablets, and mobile phones using a web browser without any sophisticated software installation.

The inclusion of visual outputs such as pictures of fishes and plants, graphs, and compatibility will enhance user interaction. The suggested system is operationally viable since it streamlines the process of maintaining an aquarium.

Legal Feasibility

Legality feasibility examines whether the proposed solution adheres to applicable laws and ethics. The AquaMind solution is legally feasible since it does not violate any laws or use any prohibited technology. The application deals with information related to aquariums and users' accounts. To ensure legal and ethical compliance, the system will:

- Protect user's account information securely.
- Avoid unauthorized sharing of personal data.
- Use licensed or open-source software technologies.
- Respect copyright rules for images and external resources.

User passwords and sensitive information will be securely stored in the database using encryption and authentication mechanisms. Any fish or plant images used in the system will either:

- Be created by the development team.
- Use royalty-free resources .
- Use properly licensed APIs or image databases.

Moreover, the solution abides by all the ethical considerations for creating software by providing correct information, proper use of artificial intelligence, and effective data processing. Consequently, the solution is legally feasible.

Feasibility Conclusion

Considering all the criteria that were analyzed during the feasibility evaluation, the AquaMind project is assessed as highly feasible. The suggested system utilizes advanced technologies which are dependable and up to date. Moreover, the development costs will not be high for the company, while the system will be easy to operate and use. It will also meet the legal and ethical requirements.

Project Description

AquaMind is an end-to-end AI-powered website aimed at supporting aquarists in managing, monitoring, and optimizing their tanks effectively and efficiently. AquaMind incorporates both the conventional aquarium management tools and the intelligent features enabled by AI to ensure a smart experience for users. The application is structured into several core modules:

1. **User Management Module**

Manages user registration, login authentication, and account management to ensure secure and personalized access to the system.

2. **Tank Management Module**

Allow users to create and manage multiple aquarium tanks, including both **monster fish tanks** and **planted tanks**, with full customization options.

3. **Fish & Plant Tracking Module**

Enables users to add, update, and manage fish and aquatic plant species within each tank for better ecosystem organization.

4. **Water Parameter Monitoring Module**

Tracks important water quality indicators such as **pH levels, temperature, and ammonia levels**, ensuring a healthy aquatic environment.

5. **Maintenance Scheduling Module**

Help users plan and manage essential tasks such as feeding, cleaning, and water changes through scheduled reminders.

6. **Fish Growth Tracking Module**

Monitors and records fish growth over time, helping users analyze development patterns and tank health.

7. **Equipment Management Module**

Manages aquarium equipment such as filters, lighting systems, and heaters to ensure proper maintenance and performance.

8. **Notification System**

Provides real-time alerts and reminders for water issues, maintenance tasks, and missed activities.

9. **Analytics Dashboard Module**

Offers visual insights and reports on tank performance, water conditions, and overall aquarium health trends.

AI-Powered and Machine learning Sub-System

AquaMind integrates advanced AI capabilities to enhance decision-making and automation:

1. **Smart Stocking Advisor**

Recommends compatible fish species based on tank size, water conditions, and species behavior.

2. **Planted Tank Assistant**

Suggests suitable aquatic plants that match tank conditions and aesthetic preferences.

3. **Water Quality Prediction System**

Uses historical data to forecast future water parameter changes and potential risks.

4. **AI Tank Designer**

Generates visual aquarium layout suggestions to help users design attractive and balanced tanks.

Overall System Overview

AquaMind incorporates both conventional aquarium management tools and AI intelligent tools to provide a full-fledged solution for aquarium enthusiasts. The solution not only helps to make management easier and risk-free but also ensures seamless user experience.

Deliverables

This project will provide a wide range of structured deliverables, which will guarantee the successful design, development, implementation, testing, and evaluation of the whole system. Every deliverable is essential for the whole life cycle of this project, providing an impact on such important aspects as functionality, usability, reliability, and maintainability of the project. The deliverables were carefully designed, ensuring their coverage of all important aspects of this system, from software development, through database design, user interaction, intelligence, documentation, and quality assurance.

1. **Fully Functional Web Application**

   A complete, responsive, and user-friendly web-based application that allows users to manage aquarium-related activities efficiently. The system will be accessible across multiple devices including desktops, tablets, and mobile phones, ensuring smooth usability and performance.

2. **User Authentication System**

   A secure authentication module that includes user registration, login, logout, password management, and profile management. This ensures that only authorized users can access the system and maintain their personal aquarium data securely.

3. Tank Management Module

A core functional module that enables users to create, update, and manage multiple aquarium tanks. It supports both monster fish tanks and planted tanks, allowing users to organize species, monitor tank conditions, and manage aquatic environments effectively.

4. Water Parameter Tracking System

A monitoring system that allows users to log and track important water quality parameters such as pH level, temperature, ammonia, nitrate, and oxygen levels. The system provides visual representations to help users understand water conditions over time.

5. AI Recommendation Engine

An intelligent recommendation system that suggests suitable fish species and aquatic plants based on tank conditions, compatibility, and user preferences. This feature enhances decision-making and helps maintain a balanced aquarium ecosystem.

6. Water Prediction Module

Predictive analytics feature forecasts future water quality conditions based on historical data. This helps users take preventive actions before issues occur, ensuring a stable and healthy aquatic environment.

7. Analytics Dashboard

An interactive dashboard that presents key insights using charts, graphs, and visual indicators. It provides an overview of tank performance, water trends, fish growth, and maintenance activities in a clear and understandable format.

8. Database Design

A well-structured relational database design including Entity-Relationship (ER) diagrams and optimized SQL schema. This ensures efficient data storage, retrieval, and scalability of the system.

9. Documentation

Comprehensive project documentation including Software Requirements Specification (SRS), system design documents, architectural diagrams, and a detailed user manual to guide users and developers.

10. Testing Reports

Detailed reports covering unit testing, integration testing, and system testing. These reports validate system functionality, identify bugs, and ensure that the application meets quality and performance standards before deployment.

Resources Required

The creation and execution of such a system will require the use of both software tools and hardware facilities. This will guarantee that everything will be done in an effective and efficient manner.

Software Resources

Software / Tool	Purpose
React	Used for developing the frontend user interface of the web application. It helps in creating a dynamic, responsive, and interactive user experience.
Node.js	Used for backend development and building RESTful APIs. It manages server-side logic and communication between frontend and database.
Python	Utilized for developing artificial intelligence and machine learning components such as recommendation and prediction models.
Microsoft SQL Server	Serves as the primary database management system for storing and managing structured data related to users, tanks, fish, plants, and water parameters.
Visual Studio Code	Acts as the main development environment (IDE) for writing, debugging, and managing project code efficiently.
Postman	Used for testing and validating APIs to ensure correct communication between frontend and backend services.
GitHub	Provides version control, collaboration, and source code management throughout the development lifecycle.

Hardware Resources

The system also needs certain hardware resources that can help in developing, testing, executing, and maintaining the software. This is important for providing computing power and connectivity during the entire process of software development.

1. **Personal computer or** laptop - Utilized for coding, testing features of the system, executing development utilities, and maintaining databases during the development process.
2. **Stable internet connection** - Needed for accessing the internet library, GitHub, various APIs, and cloud-based services.
3. **Modern web browser** - Used for deploying, testing, and debugging the web app to ensure its compatibility with various operating systems and hardware configurations.
4. **Storage devices or cloud storage** - Needed for backing up files, data sets, database dumps, source codes, and documentation of the project.

Expected Output and Outcome

Expected Outputs

The system should be able to provide a full-fledged, functional artificial intelligence-driven aquarium management web application that will generate the following outputs:

1. An integrated, web-based solution that enables effective management of all the tasks associated with an aquarium.
2. AI-driven recommendation engines that will intelligently recommend fishes and plants suited to tank conditions.
3. Modules that will help predict future trends in terms of water quality.
4. Interactive dashboards displaying real-time insights, charts, and performance reports related to aquarium health.

Expected Outcome

The adoption of the system will lead to major advancements in aquarium management, such as:

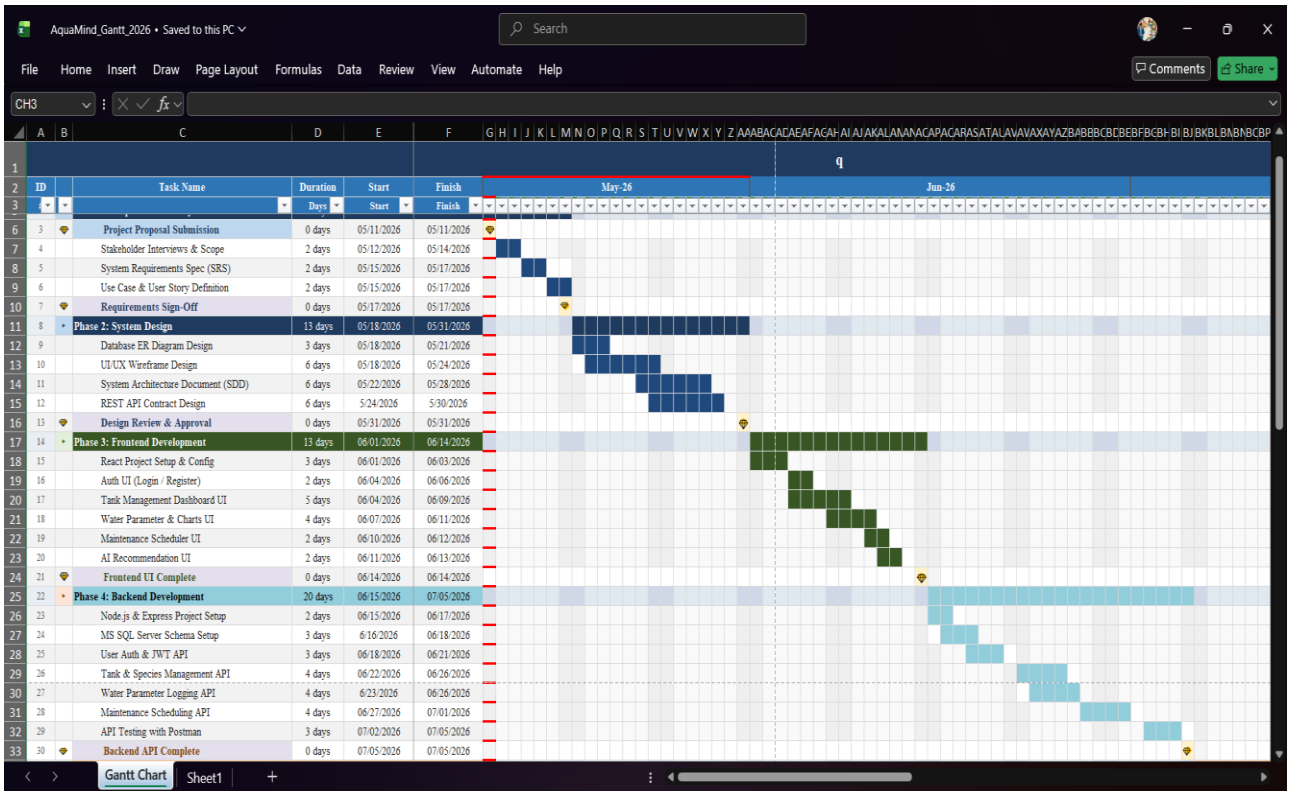
1. Improved health management of aquariums due to constant monitoring of the environment and tank condition.
2. A decrease in mortality rates among fish through timely warning messages and suggestions from the system.
3. Intelligent suggestions from the system that allow making informed decisions.
4. Facilitated maintenance and monitoring of aquariums for aquarium enthusiasts.
5. In general, the system will help achieve a more efficient and automated process of aquarium management.

Time Plan for Implementation

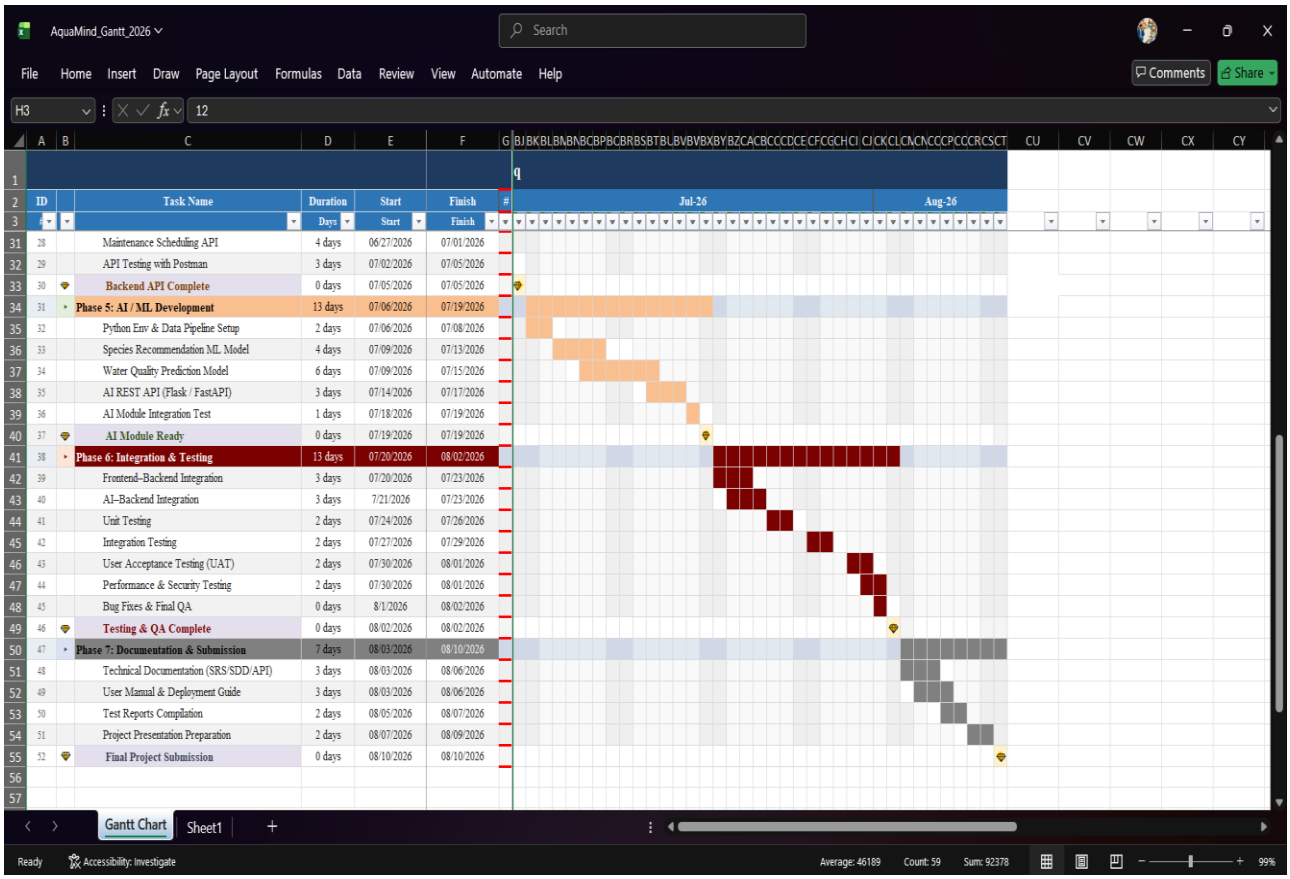
This project will be conducted through various phases to make sure that there is an efficient and systematic process of developing this system. The system development process includes planning, designing, developing, testing, and documenting. Every phase will be devoted to one aspect of the system development process.

Phase	Activity	Duration
1	Requirement Analysis	Week 1
2	System Design	Weeks 2–3
3	Frontend Development	Weeks 4–5
4	Backend Development	Weeks 6–8
5	AI Development	Weeks 9–10
6	Integration & Testing	Weeks 11–12
7	Documentation	Week 13

AquaMind - Grantt Chart



ID	Task Name	Duration	Start	Finish
		Days	Start	Finish
3	Project Proposal Submission	0 days	05/11/2026	05/11/2026
4	Stakeholder Interviews & Scope	2 days	05/12/2026	05/14/2026
5	System Requirements Spec (SRS)	2 days	05/15/2026	05/17/2026
6	Use Case & User Story Definition	2 days	05/15/2026	05/17/2026
7	Requirements Sign-Off	0 days	05/17/2026	05/17/2026
8	Phase 2: System Design	13 days	05/18/2026	05/31/2026
9	Database ER Diagram Design	3 days	05/18/2026	05/21/2026
10	UI/UX Wireframe Design	6 days	05/18/2026	05/24/2026
11	System Architecture Document (SDD)	6 days	05/22/2026	05/28/2026
12	REST API Contract Design	6 days	5/24/2026	5/30/2026
13	Design Review & Approval	0 days	05/31/2026	05/31/2026
14	Phase 3: Frontend Development	13 days	06/01/2026	06/14/2026
15	React Project Setup & Config	3 days	06/01/2026	06/03/2026
16	Auth UI (Login / Register)	2 days	06/04/2026	06/06/2026
17	Tank Management Dashboard UI	5 days	06/04/2026	06/09/2026
18	Water Parameter & Charts UI	4 days	06/07/2026	06/11/2026
19	Maintenance Scheduler UI	2 days	06/10/2026	06/12/2026
20	AI Recommendation UI	2 days	06/11/2026	06/13/2026
21	Frontend UI Complete	0 days	06/14/2026	06/14/2026
22	Phase 4: Backend Development	20 days	06/15/2026	07/05/2026
23	Node.js & Express Project Setup	2 days	06/15/2026	06/17/2026
24	MS SQL Server Schema Setup	3 days	6/16/2026	06/18/2026
25	User Auth & JWT API	3 days	06/18/2026	06/21/2026
26	Tank & Species Management API	4 days	06/22/2026	06/26/2026
27	Water Parameter Logging API	4 days	6/23/2026	06/26/2026
28	Maintenance Scheduling API	4 days	06/27/2026	07/01/2026
29	API Testing with Postman	3 days	07/02/2026	07/05/2026
30	Backend API Complete	0 days	07/05/2026	07/05/2026



ID	Task Name	Duration	Start	Finish
		Days	Start	Finish
28	Maintenance Scheduling API	4 days	06/27/2026	07/01/2026
29	API Testing with Postman	3 days	07/02/2026	07/05/2026
30	Backend API Complete	0 days	07/05/2026	07/05/2026
31	Phase 5: AI / ML Development	13 days	07/06/2026	07/19/2026
32	Python Env & Data Pipeline Setup	2 days	07/06/2026	07/08/2026
33	Species Recommendation ML Model	4 days	07/09/2026	07/13/2026
34	Water Quality Prediction Model	6 days	07/09/2026	07/15/2026
35	AI REST API (Flask / FastAPI)	3 days	07/14/2026	07/17/2026
36	AI Module Integration Test	1 days	07/18/2026	07/19/2026
37	AI Module Ready	0 days	07/19/2026	07/19/2026
38	Phase 6: Integration & Testing	13 days	07/20/2026	08/02/2026
39	Frontend–Backend Integration	3 days	07/20/2026	07/23/2026
40	AI–Backend Integration	3 days	7/21/2026	07/23/2026
41	Unit Testing	2 days	07/24/2026	07/26/2026
42	Integration Testing	2 days	07/27/2026	07/29/2026
43	User Acceptance Testing (UAT)	2 days	07/30/2026	08/01/2026
44	Performance & Security Testing	2 days	07/30/2026	08/01/2026
45	Bug Fixes & Final QA	0 days	8/1/2026	08/02/2026
46	Testing & QA Complete	0 days	08/02/2026	08/02/2026
47	Phase 7: Documentation & Submission	7 days	08/03/2026	08/10/2026
48	Technical Documentation (SRS/SDD/API)	3 days	08/03/2026	08/06/2026
49	User Manual & Deployment Guide	3 days	08/03/2026	08/06/2026
50	Test Reports Compilation	2 days	08/05/2026	08/07/2026
51	Project Presentation Preparation	2 days	08/07/2026	08/09/2026
52	Final Project Submission	0 days	08/10/2026	08/10/2026

AquaMind Milestone

Milestone	Date
Project Proposal Submission	11 May 2026
Requirements Sign-Off	17 May 2026
Design Review & Approval	31 May 2026
Frontend UI Complete	14 Jun 2026
Backend API Complete	05 Jul 2026
AI Module Ready	19 Jul 2026
Testing & QA Complete	02 Aug 2026
Final Project Submission	10 Aug 2026

Limitations

While the system incorporates sophisticated functions, automation, and intelligence, some shortcomings can be anticipated in the first edition owing to the technical, material, and developmental challenges involved. Some limitations may impact the functioning of some features and the performance of the system itself.

1. **AI accuracy depends on data quality.**

   The functioning of the recommendation and prediction algorithms by AI is highly dependent on the correctness of information provided by users. Inaccurate data can negatively impact the efficiency of the system.

2. **No real-time IoT sensor integration.**

   Now, there are no functions that allow linking to hardware sensors. Users should manually enter water parameters.

3. **Internet connection required.**

   Given that the platform operates via a web browser, having an active internet connection is a requirement.

4. **Limited initial training dataset.**

   The accuracy of machine learning models will decrease due to the small amount of initial data available; however, with time, it will increase.

5. **No native mobile application in version one.**

   The initial release of the system will work as a website. It is planned to introduce mobile applications in later releases.

Conclusion

In summary, AquaMind is expected to revolutionize aquarium management and offer intelligent and effective solutions that will make life easier for aquarium enthusiasts. The system takes care of a variety of usual challenges, from water quality control, maintenance schedule, species compatibility, and even aquarium health monitoring. Using Artificial Intelligence (AI), Machine Learning (ML), and advanced web technology, the system provides innovative features like fish and plant recommendation system, water quality predictor, intelligent alerts and notification, and interactive dashboards. All these innovations will assist users in making sound decisions and maintaining healthy aquatic life conditions.

From the feasibility assessment results, it is evident that the project is technically feasible, economically viable, operationally realistic, and legally permissible. Moreover, the chosen technologies will guarantee smooth development of AquaMind system. Such technologies include React, Node.js, Python, and Microsoft SQL Server. Overall, AquaMind system will be able to significantly change the way traditional aquarium management works for aquarium hobbyists.

References

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